17.353 1+0.0000 1+0.00 11 119.553 1+0.0000 1+0.00 11

Exp A: per-corpus calibration baseline
the streaming filter is not online MLE

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1 Question

The class proposal's headline framing positions the streaming Laplace $\hat{\beta}$ filter as "online temperature scaling [Guo17] converging in the large-data limit to the same global T Guo's batch MLE fits." This experiment tests that claim directly.

For each of WikiText-103 and WritingPrompts, we (i) fit two $\hat{\beta}$ estimators on a calibration half—streaming ($\hat{\beta}_{\text{stream}}$, per-passage filter, aggregated by median, and a pooled single-trajectory variant) and Guo MLE ($\hat{\beta}_{\text{MLE}}$, scipy on the calibration-half NLL), and (ii) evaluate the per-token test PPL under each.

2 Setup

Qwen-2.5-3B, no fine-tuning. 60 calibration passages plus 60 test passages per corpus, each tokenized to exactly 256 tokens. Passage-level (not token-level) split: passages 0–60 are calibration, passages 60–2 60 are test. Filter: $\sigma_0 = 0.5$, warmup 20 tokens, 236 filter steps per passage.

Calibration estimators:

$\hat{\beta}_{\text{stream\_med}}$. Run the existing Laplace filter (`counter_decode.laplace_update`) per-passage with state reset between passages; take the median final $\hat{\eta}$ across passages.

$\hat{\beta}_{\text{stream\_pooled}}$. One filter trajectory across all $60 \cdot 236 =\approx 14{,}160$ calibration tokens, no resets. This is the apples-to-apples object vs. a batch fit.

$\hat{\beta}_{\text{MLE}}$. `scipy.optimize.minimize_scalar` on the per-token NLL $-\frac{1}{T}\sum_t \log \operatorname{softmax}(\beta \cdot \ell_t)[x_{t+1}]$ over the same calibration tokens.

Test evaluation: per-passage per-token NLL on the test split at four β candidates (T = 1, own-corpus streaming, own-corpus MLE, *other*-corpus MLE). Headline statistic is per-passage $\Delta \log \text{PPL} = \log \text{PPL}(\beta) - \log \text{PPL}(1)$, with paired t-tests against zero across the 60 test passages.

3 Headline

Three structural facts.

Table 1: Calibration $\hat{\beta}$ values per estimator. All as $\log\hat{\beta}$; $\log\hat{\beta} = 0 \Leftrightarrow \hat{\beta} = 1$.

corpus	$\log\hat{\beta}_{\text{stream\_med}}$	$\log\hat{\beta}_{\text{stream\_pooled}}$	$\log\hat{\beta}_{\text{MLE}}$
WikiText	-0.059	-0.937	+0.000
WritingPrompts	-0.314	-0.802	-0.036

Table 2: Test PPL by β setting (per-token, averaged over passages). $\Delta\log\text{PPL}$ is per-passage paired against the $T = 1$ baseline; t is the one-sample t-stat. Significance after Bonferroni correction ($k = 8$) requires $|t| > 2.85$.

test corpus / β setting	$\hat{\beta}$ value	test PPL	$\Delta\log\text{PPL}$	paired t
WikiText / $T = 1$	1.000	1	1	1
WikiText / stream_med		7.385	+0.0048	+4.96
WikiText / stream_pooled		105.248	+2.6543	+160.77
WikiText / WT_MLE		7.353	+0.0000	+1.86
WikiText / WP_MLE		7.361	+0.0014	+2.31
WritingPrompts / $T = 1$	1.000	1	1	1
WritingPrompts / stream_med		24.989	+0.2522	+26.46
WritingPrompts / stream_pooled		143.398	+2.0103	+89.13
WritingPrompts / WP_MLE		19.485	-0.0011	-0.95
WritingPrompts / WT_MLE		19.555	+0.0001	+3.96

(a) The streaming filter is not online MLE. Calibration NLL at $\hat{\beta} = 1$ vs at $\hat{\beta}_{\text{MLE}}$ differs in the fourth decimal place on both corpora (1.9724 vs 1.9724 on WikiText; 3.0029 vs 2.9992 on WritingPrompts). This places the Guo MLE essentially at $\log\beta = 0$. The pooled streaming filter on the *same data* settles at $\log\hat{\beta} = -0.937$ (WikiText) and -0.802 (WritingPrompts) — gaps of ≈ -0.938 and -0.765 nats from MLE. The two estimators have different fixed points.

(b) Why the gap. The filter's per-step update (`counter_decode.py:91--182`) computes $q = \text{softmax}(\ell_t)$, $E_q[\ell_t]$, $\text{Var}_q[\ell_t]$ — all at $\beta = 1$, never reevaluated at the running $\hat{\beta}$. This is by design: the matched-prior bootstrap-reference framework (`bayesian-sampler/counter_decoding.html`) defines $\ell_t^* = \ell_t/\hat{\beta}$ exactly so that the score and Fisher are $\hat{\beta}$-independent. The filter is therefore a per-token Bayesian update *under that imprint model*, not an iterated Newton solver on the Guo NLL. Its pooled final $\hat{\eta}$ behaves like a single damped Newton step from the prior mode using $\sum_t\big(\ell_t[x_{t+1}] - E_q[\ell_t]\big)$ in the numerator and $1/\sigma_0^2 + \sum_t \text{Var}_q[\ell_t]$ in the denominator — accurate near $\beta = 1$, but increasingly biased as one walks away from $\beta = 1$ on a non-quadratic NLL. Figure 1 shows exactly this geometry: the NLL surface is flat near the minimum, so a small gradient at $\beta = 1$ divided by a smaller curvature lands far away.

(c) Applied as a calibrator, the filter hurts test PPL. On WritingPrompts test, $\hat{\beta}_{\text{stream\_med}}$ (a per-passage estimator close to what one would actually apply online) increases per-token PPL from 1 to 24.989 — $\Delta\log\text{PPL} = +0.2522$ ($t = +26.46$). The pooled streaming $\hat{\beta}$ is catastrophic. $\hat{\beta}_{\text{MLE}}$ is statistically indistinguishable from $T = 1$ (paired $t = -0.95$, $p = 0.347$). Cross-corpus MLE (WT $\rightarrow$ WP and WP $\rightarrow$ WT) is also indistinguishable from $T = 1$.

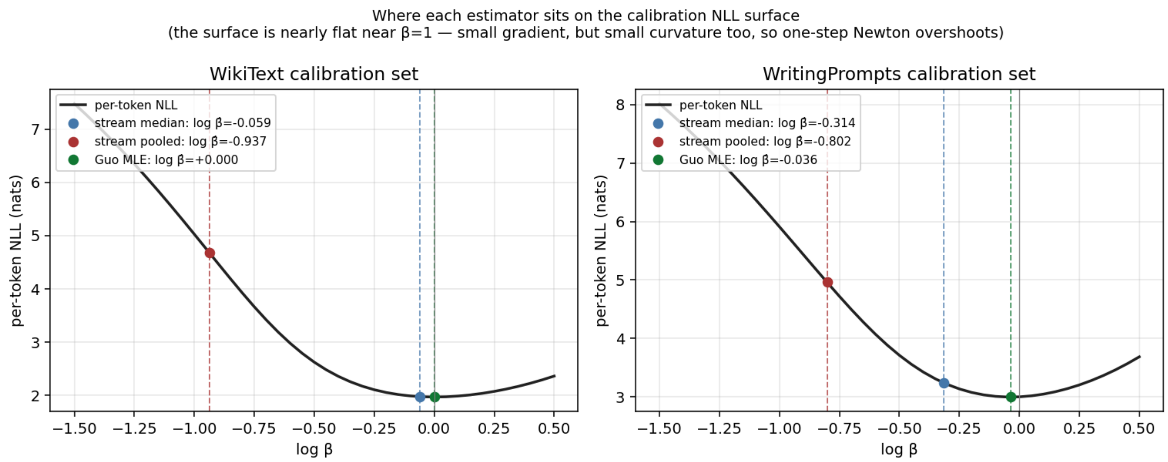

Figure 1: Per-token calibration NLL as a function of $\log \beta$, with each estimator marked. The minimum is at $\log \beta \approx 0$ on both corpora (green) — Qwen-2.5-3B is essentially well-calibrated. The pooled streaming filter (red) lands far up the steep wall of the NLL surface, not at the minimum. The per-passage-median streaming (blue) is closer to MLE but still biased downward on WritingPrompts.

4 What this implies for the project

1. **The streaming filter is not online temperature scaling.** It measures the model's local belief about its decoder under the matched-prior imprint hypothesis. The Guo MLE measures the post-hoc temperature that maximizes target-token likelihood. On Qwen-2.5-3B the two place $\hat{\beta}$ at $\log \beta = -0.94$ and 0.00 respectively on WikiText — not consistent in the large-data limit. The proposal's claim of asymptotic equivalence (`decoding-modeling.tex`) does not survive this test and should be removed.

2. **Qwen-2.5-3B is essentially well-calibrated** on these corpora. The Guo NLL minimum sits at $\log \beta \approx 0$. There is no meaningful post-hoc temperature gain to be had from either WikiText or WritingPrompts text in this length/model regime. Any future Exp B (per-context sliding $\hat{\beta}_t$) cannot defeat $\beta = 1$ on calibration grounds alone.

3. **The filter still has an interpretable measurement target**: the model's gradient signal at $\beta = 1$, which under the imprint hypothesis is the model's local belief about how much its prefix was temperature-scaled. This is a different research question than calibration; the project's framing should track that. In particular, Exp C (entropy after a $\beta_{\text{dec}} \neq 1$ prefix) directly probes whether the model's distribution responds to the apparent decoder of its own context.

4. **The original natural-text findings retain their narrow content**: the per-passage filter showed no detectable baseline drift under self-sampling (`natural_text/findings.pdf`). This was a filter-pathology check, not a calibration claim, and is unaffected.

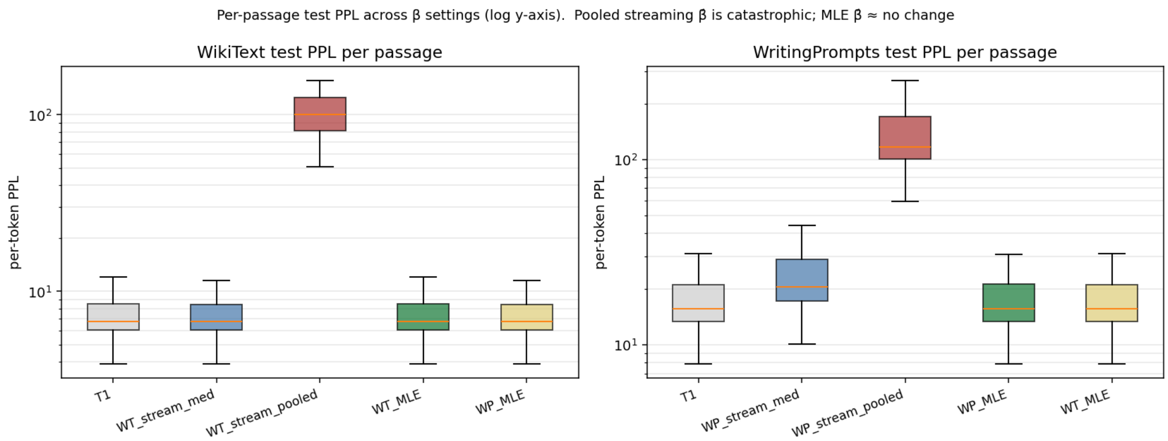

Figure 2: Per-passage test PPL across β settings (log y-axis). Applying $\hat{\beta}_{\text{stream\_pooled}}$ at test time costs 14× (WikiText) and 7× (WritingPrompts) increases in PPL. $\hat{\beta}_{\text{MLE}}$ is statistically indistinguishable from $T = 1$.

5 Limitations

- One model (Qwen-2.5-3B), one tokenization, one passage length (256). The MLE-vs-streaming gap and the "Qwen is well-calibrated" finding both depend on these.
- No held-out validation set within the calibration half. All calibration-MLE fitting is on the same tokens used to compute the streaming $\hat{\beta}$. Test PPL is on disjoint passages, so the headline test result is properly held-out, but per-corpus $\hat{\beta}$ comparisons are in-sample.
- Per-passage and pooled streaming differ in aggregation; the per-passage median is the operational quantity (what one would compute online), the pooled trajectory is the apples-to-apples comparison vs. pooled MLE. Both disagree with MLE. Other aggregations (mean, precision-weighted mean) were not explored.

6 Bottom line

- Streaming filter $\neq$ Guo MLE on real data. Different objectives, different fixed points, large gap (≈ 1 nat in $\log \hat{\beta}$).
- Qwen-2.5-3B is well-calibrated on these corpora; no post-hoc temperature gain is available.
- Applying the streaming $\hat{\beta}$ as a calibrator increases test PPL.
- The filter still measures something coherent — the model's local belief about its decoder — but the project's framing should treat "calibration" and "imprint detection" as distinct.